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Return to Play and Re-Injury Rates in NCAA Division I Football Players Following Anterior Cruciate Ligament Reconstruction Using Hamstring Autograft

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High Return to Play and Low Re-Injury Rates in NCAA Division I Football Players Following Anterior Cruciate Ligament Reconstruction Using Quadrupled Hamstring Autograft

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High Return to Play and Low Re-Injury Rates in NCAA Division I Football Players Following Anterior Cruciate Ligament Reconstruction Using Quadrupled Hamstring Autograft

Purpose: The purpose of this study was to examine the outcomes of ACL reconstruction using quadrupled hamstring (QH) autograft in a cohort of National Collegiate Athletic Association (NCAA) Division I football players.

Methods: A retrospective analysis was performed on NCAA Division I football players at a single institution who had transtibial ACL reconstruction using QH autograft from 2001 to 2016. Primary outcomes were ACL re-injury and return to play (RTP). Secondary outcomes were position, percent eligibility used after surgery, graft diameter, Tegner Lysholm scores, concomitant injuries/surgeries, and post-collegiate professional play.

Results: Thirty-four players had QH autograft ACL reconstruction from 2001-2016. 29 players achieved RTP. Two (6.9%) of the 29 sustained ACL re-injuries. The average RTP was 318 days (range 115-628) after surgery. Players used 79.5% of their remaining collegiate eligibility after surgery. Nine players sustained multi-ligamentous knee injuries. This did not have a significant effect on RTP (p=0.709. mean 306$\pm$24 days for isolated ACL, mean of 353$\pm$51 for 2 ligaments, mean of 324$\pm$114 for 3+ ligaments) and none sustained re-injury. Twenty-eight sustained associated meniscal injury and 8 sustained chondral injuries. The mean post-operative Tegner-Lysholm score was 90.7 out of 100 with mean follow-up of 102 months. Eighteen of these players went on to play professionally, with 17 joining National Football League (NFL) rosters and one an arena team roster.

Conclusion: QH demonstrated an ACL re-injury rate and RTP similar to previously published predominantly BTB ACL re-injury data in elite athletes. This study demonstrates that QH autograft may be a viable option in elite athletes.

Level of Evidence: IV – Case Series

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Introduction

Anterior cruciate ligament (ACL) injury in elite football players is a common and devastating injury. The goal of surgical intervention on ACL injuries is to restore stability of the knee and allow players to return to play (RTP) at their previous level of performance. From 1987 to 2000, 8% of all 5047 players attending the NFL combine had a history of ACL injury and 6% had a subsequent reconstruction.[1] According to Dragoo et al, the overall incidence rate of ACL injuries in NCAA football players from 2004-2009 was 1.42 per 10,000 games or practices.[2] A recent study by Daruwalla et al. reports that about 20% of NCAA players fail to return to sport after having an ACL reconstruction.[3] These authors found that players whose ACL was reconstructed with autograft returned to play more often than those whose ACL was reconstructed with allograft. They did not find any difference between the types of autograft used for reconstruction, however 90% of the grafts were BTB vs only 10% QH.

Traditionally, bone-patella tendon-bone (BTB) grafts have been the gold standard of graft choice in elite athletes. In a 2008 survey, Shrock et al found that NFL team physicians reported using BTB in 87% of their ACL reconstructions. That reported preference rose to 97% in 2016. Zero out of 29 reported using hamstring autograft.[4] Of 49 ACL tears occurring in NFL athletes from 2001-2006, 47 were reconstructed using BTB and 2 using QH.[5] Carver et al surveyed 100 NCAA Division I team physicians which revealed a similar trend with 21% of surgeons using QH in 2008 and only 12% in 2016. With regards to BTB, the usage was 67% and 83% in 2008 and 2016, respectively.[6] Despite this trend, a paucity of evidence exists showing BTB is a superior option when compared to quadrupled hamstring (QH) autograft in elite athletes and more specifically American football players. With regards to all comers and not just elite football players, Kaeding et al studied 2488 patients and demonstrated a 4.6% reinjury rate with QH graft at 2 year follow-up compared to 3.2% reinjury with BTB[7]. Two very large 2014 studies from Scandinavia and Norway of 45,998 and 12,643 patients, respectively, found a statistically significant lower risk of revision with BTB grafts vs QH[8,9]. These studies may contribute to the increased trend of BTB being the most oft-used graft in elite athletes.

A graft that can withstand the rigors of elite football performance is of utmost importance. Graft rupture is not uncommon, with 12.3% of all NFL ACL injuries occurring in previously reconstructed knees.[10] Lai et al

performed a meta-analysis of graft re-injury rate in elite athletes including football players undergoing ACL reconstruction and found a rate of 5.2%.[11] The ultimate timing of return to play (RTP) is defined by a period of time to allow the graft to heal along with an assessment of the athlete's strength, proprioception and kinematics. Shrock et al found that the most common time NFL team physicians allowed RTP was 7 to 9 months. In this study, only 14% of surgeons allowed a return before 6 months.[4] While both QH and BTB require adequate time to undergo "ligamentization" (a process where the graft initially breaks down and then remodels into a ligament with creeping substitution[12]), a soft tissue only graft like QH also needs to adhere to the femoral and tibial tunnels by forming a fibrovascular scar. It has been found that soft tissue grafts incorporate into their bony tunnels more slowly than BTB.[13] For this reason, most surgeons support a BTB graft in elite football players as it allows for bone to bone healing.

A major concern with QH is the donor site morbidity and potential loss of knee flexion strength, particularly in competitive athletes. Cristiani et al examined the post-operative limb symmetry index (LSI) of patients who were randomized to ACL reconstruction with either BTB or QH. They found a significant decrease in hamstring strength in the QH (89.1% LSI) group vs BTB (97.1% LSI) at 24 months. Conversely, they found no significant difference in quadriceps strength in the BTB group vs QH at 24 months.[14] Proponents of QH as a graft choice will cite its superior ultimate load to failure versus BTB (4590N vs 2977N), as well as lower incidence of anterior knee pain and stiffness, without risk of patella fracture.[15]

The purpose of this study was to examine the outcomes of ACL reconstruction using quadrupled hamstring (QH) autograft in a cohort of National Collegiate Athletic Association (NCAA) Division I football players. We hypothesized that the results of this study in a cohort of athletes who had QH reconstruction are similar to historical numbers for athletes undergoing predominantly BTB ACL reconstruction.

Methods

Institutional review board approval was obtained for this study and all athletes provided informed consent prior to data analysis allowing access to their prior records as well as a survey of their current functional status. Medical records of all male varsity football players at the institution from 2001 to 2016 were retrospectively reviewed. All

players who sustained an ACL tear and underwent subsequent reconstruction while on the football roster were included. All ACL reconstructions were performed by the head orthopedic surgeon and all players in this study received QH autograft. No players required allograft augmentation. All players underwent a standardized post-operative rehabilitation protocol led by the athletic training staff with modifications made as prescribed by the surgeon. Players who had prior ACL reconstruction, those in which a full set of medical records could not be obtained, and those who sought initial surgical treatment elsewhere were excluded. The primary endpoints were re-injury and RTP. Secondary endpoints included: percentage of NCAA eligibility used after injury, player position, associated injuries of the ipsilateral knee, need for additional surgery on the ipsilateral knee, subjective sensation of pain or instability measured by the Tegner Lysholm knee score, and progression to the National Football League (NFL) after RTP.

Data were obtained from three sources as follows: player medical records, institutional medical database, and player survey. All player medical records were stored with the athletic training department in a secure custom database. Data collected from the player medical record included age, height, weight, body mass index (BMI), year of injury, date of surgery, graft diameter, concomitant knee injuries, ipsilateral knee surgeries, and ACL re-injury. Player participation data are stored in the secure database within the institution's sports information department. Data collected from the sports information database included eligibility year at the time of injury, position, date of return to game, number of games played after return, and number of remaining games of eligibility. A standardized Qualtrics (Provo, Utah) questionnaire was sent to each player by a secure link through email. Data collected from the questionnaire included consent to participate in the study, the Tegner Lysholm knee scoring scale, and any history of ipsilateral knee surgery after leaving the institution. NFL team participation was confirmed via each team's media report and website. Statistical analysis was performed using JMP 15.0 (SAS, Inc., Cary, NC) to run T-tests.

The decision to formally clear a player for return to sport was determined by collaborative decision involving the football athletic training staff and the head orthopedic surgeon. Criteria used to clear a player for sport included: isokinetic muscle testing equal or greater than 80% of baseline strength testing on contralateral side at 60 degrees per second (Cybex Norm testing), passing of functional testing progression in rehabilitation for

position specific drills, passing of non-contact modified team practice, subjective assessment of mental readiness of the player, favorable physical exam by surgeon. Time from index surgery was not used as criteria.

Return to play was defined as actual game play in an official NCAA or professional game after ACL reconstruction. Eligibility remaining was determined based on the number of potential games a player can play in after surgery during a 4 year tenure, starting with the beginning of the following season. For example, if a player sustained an ACL rupture in the midseason of his freshman year, the potential games remaining would have been all games in his sophomore through senior year. The percent eligibility used was then calculated by the total number of games played after injury divided by potential games remaining. This calculation excluded redshirt years. If a player chose to participate in a redshirt year, the games missed that year would not be factored in, only the games remaining in the non-redshirt seasons.

All procedures were performed by the head team surgeon. The ACL reconstruction surgical technique for each was identical for each patient over the course of this study. All patients had routine diagnostic arthroscopy and meniscal/chondral pathology was addressed followed by ACL reconstruction. In each patient, both the semitendinosus and gracilis were harvested. The tendons were then prepped by stripping all muscle and tubularizing the musculotendinous end of the graft. The free ends were then whip-stitched and each tendon was doubled to create a quadrupled graft. In no cases was either tendon tripled, nor was there any allograft augmentation. Had any grafts been below 8mm, a gracilis allograft would have been used for augmentation. The tibial tunnel was drilled using a tibial guide placed through the anteromedial portal. The femoral tunnel was then drilled using a trans-tibial technique with the knee in 80 degrees of flexion. The tunnel size was matched to the diameter of the graft. The graft was then passed in a retrograde fashion through both tunnels using a proprietary industry device and then fixated on the femoral side using a Bio-Transfix™ (Athrex; Naples, FL) bioabsorbable crosspin device. This device is preferred by the senior surgeon due to a high load-to-failure as well as minimal graft elongation[16]. The tibial side was then secured using a BioComposite Interference Screw (Arthrex; Naples, FL).

Results

In total, 35 players sustained ACL injuries between 2001 - 2016. One was excluded as full medical records could not be obtained, leaving 34 players included in this study. The most common position to sustain an ACL tear was running back with a total of 8 players. The most common academic year to sustain this injury was junior year with 10. Table 1 demonstrates a breakdown of demographical data.

Demographical Data	
	N
Position	
Running Back	8
Defensive Line	5
Linebacker	5
Offensive Line	5
Quarterback	3
Wide Receiver	3
Long Snapper	3
Cornerback	2
Fullback	1
Year	
Freshman	9
Sophomore	9
Junior	10
Senior	6

Table 1 – Position and year of eligibility while the injury was sustained.

Isolated ACL injuries were sustained in 25 of the 34 players. Of the remaining 9 players, a breakdown of their injuries and treatments can be seen in Table 2.

Multiligamentous Injuries		
# of Players	Injured Ligament	Treatment
5	MCL	nonoperative
1	MCL	MCL repair
1	MCL + LCL	nonoperative
1	MCL + PCL	MCL repair
1	MCL + LCL + PCL	PLC reconstruction (allograft)

Table 2. Number and characteristics of multiligament injuries with their treatments.

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An associated meniscus injury was noted in 28 of the 34 players. The injuries and treatments are broken down in Table 3.

Associated Meniscal Injuries		
# of Players	Associated Injury	Treatment
16	M or L Meniscus	Partial Menisectomy
4	M or L Meniscus	Unknown
2	M or L Meniscus	Repair
6	M and L Meniscus	Repair

Table 3. Number and description of concomitant meniscal injuries with their treatments.

Eight of 34 players were noted to have a chondral injury. Six of these injuries required surgical intervention during their index procedure. Of the 6, 4 players underwent chondroplasty and 2 underwent microfracture.

Return to Play (RTP)

Return to play at any level (professional or collegiate) following surgery was accomplished by 29 of the 34 players. Two of the 5 players who did not RTP were seniors who did not go on to professional play. One player decided to forgo his senior year and signed as an undrafted free agent in the NFL the following season but was released without playing in a game. The other two players included one who decided to no longer play football and another who never gained medical clearance. Of the 29 players who did return, the average time after surgery for RTP was 318 days ± 111 days (mean ± SD). The fastest RTP was 115 days and the slowest was 628 days. Twenty-three of the 29 players RTP in less than 1 year after surgery. The player who RTP at day 628 initially underwent medial and lateral partial meniscectomies as well as microfracture of the medial femoral condyle for an osteochondral defect. During his recovery period, he sustained another meniscal injury requiring repeat partial meniscectomy of both the medial and lateral menisci.

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ACL Re-injury and Subsequent Surgeries

Re-injury occurred in 2 out of the 29 (6.9%) players that returned to play. These players were both running backs. Both players sustained traumatic re-injuries with direct-blow contact injuries to their operative knee. One re-injury occurred at 7 months post-operatively and the other was 5 years after the index procedure. Notably, both had lateral meniscus injuries necessitating debridement at the time of the index procedure as well as at re-injury. These players both underwent revision ACL reconstruction with BTB autograft at this institution with the same primary surgeon. One other player sustained an ACL tear to the contralateral knee. There were 8 other subsequent ipsilateral knee surgeries in 7 patients which included one hematoma evacuation, three chondroplasties, five partial menisectomies. The timing of these ranged from 2 weeks to 3 years post-operatively. In total, 9/34 (26.5%) players required a second procedure on their ipsilateral knee.

Graft Diameter

All grafts were QH autograft. Graft diameter ranged from 8.0mm to 9.5mm (table 4). One operative note failed to mention graft size. The two players who sustained re-injury had a graft diameter of 8.0mm and 9.5mm, respectively.

ACL Graft Sizes	
Graph Size (mm)	n = 34
9.5	5 (15%)
9	11 (32%)
8.5	10 (29%)
8	7 (21%)
Unknown	1 (3%)

Table 4. Number of players receiving QH autograft according to graft size.

Remaining Eligibility Used

Of the 34 players, 30 had NCAA eligibility remaining after sustaining injury. The others were seniors without the ability to obtain a redshirt year. Of these 30 players, 25 (83%) used some or all of their remaining eligibility. The 5 players who did not return to play included one player that chose to no longer participate in college athletics, one that sustained additional injuries and never gained medical clearance, and one that left college early to pursue

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a career in the National Football League. The remaining 2 transferred and research attempts failed to obtain accurate records of their future game participation. These 25 players with remaining eligibility averaged a return to 19.8 games of a possible 24.9 games using 79.5% of their remaining eligibility.

Multiligament Injuries

There were 9 players that sustained injuries to two or more ligaments. Six of them returned the following season to competitive football at either the collegiate or professional level. Four of these players with two or more ligaments injured returned to the study institution and used 85 of their 115 games eligible, or 73.9% of remaining eligibility. This was in contrast to players with isolated ACL injuries who used 80.7% of their remaining eligibility. This difference did not reach statistical significance ($p = 0.368$). The average RTP for these players with multiligamentous injuries was 348 days vs 273 for isolated ACL injuries, but this difference did not reach statistical significance ($p = 0.709$).

Survey Results

A post-operative Tegner-Lysholm survey was completed by 33 out of 34 players. Mean time between surgical intervention and completion of the questionnaire was 102 months ± 52 months (mean ± SD, range 24 to 198 months). All players had at least 2 years of post-operative recovery prior to completing their survey. The average score was 90.7 ±14.2 out of a possible 100 (range 43 to 100). Seventeen of the 33 respondents scored 100. Table 5 demonstrates the breakdown by overall score category. The most common symptoms reported were pain and swelling, with 14 and 12 players reporting some level of each, respectively.

Tegner Lysholm Knee Survey Scores	
Score	n = 33
91-100 (Excellent)	22 (67%)
84-90 (Good)	3 (9%)
66-83 (Fair)	5 (15%)
0-65 (Poor)	3 (9%)

Table 5. Tegner-Lysholm survery outcomes for players by score category.

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Professional Careers

Seventeen of the 34 players eventually went on to join an NFL roster. Seven of these made the active roster, three made the practice squad, six were invited to training camp and one was invited to rookie mini-camp. One went on to play in the arena league.

Discussion

This study found an ACL re-injury rate of 6.9%. Additionally, this study demonstrated 85.3% of players returned to NCAA or NFL play the following season after ACL reconstruction. Lai et al. showed that 78% of elite American professional or NCAA Division I football players returned to sport following ACL reconstruction.[11] Shah et al found that 63% of NFL athletes who sustained an ACL injury during their career returned to NFL game play. Of their cohort, 4% had QH grafts and 96% had BTB.[5] McCullough et al demonstrated in their MOON cohort that 63% of high school football players and 69% of collegiate football players returned to play with mixed graft choices (BTB, QH, allograft).[17] The results of the current study show that elite football players with ACL reconstruction using QH graft return to play at a rate comparable to that reported in the literature for predominantly BTB grafts (table 6).

Study	N	Population	Focus	Findings	Comments
Daruwalla et al[3]	184	Division 1 Football Players	Return to Play	82% RTP 182 had graft information: 140 BTB, 15 QH, 27 Allo.	Higher % RTP with QH, but not statistically significant
Lai et al[11]	1272	Professional or Division 1 Athletes	Return to Sport and Graft Rupture	78% of players returned to sport	Graft rupture rate of 5.2%.
Brophy et al[1]	94	NFL	Effect of Isolated Menisectomy, Isolated ACL reconstruction, and both combined on career length	Isolated ACL reconstruction had no effect on career length. Isolated menisectomy and combined shortened careers.	6% of NFL players had a subsequent revision reconstruction for re-tear
Shah et al[5]	49	NFL	Return to Sport	63% of NFL athletes who sustained an ACL injury during their career returned to NFL game play	4% had QH grafts and 96% had BTB
McCullough et al[17]	147	High School and College Football Players (MOON)	Return to Play	63% of high school football players and 69% of collegiate football players returned to play	mixed graft choices (BTB, QH, allograft) but BTB most

					common
Kamath et al[18]	89	NCAA Division 1 Athletes	Comparison of Graft Survivorship, Reinjury Rates, Eligibility Used in players who had surgery before college (PC) versus during (IC)	1 ACL re-injury of 54 (1.9%) in their IC group and 6/35 (17.1%) in their PC group PC used 78% remaining eligibility, IC used 77%.	All PC grafts that failed were BTB

Table 6 – Comparison of Studies Performed on Elite Athletes

There are several strengths in this study. This is a large cohort of elite football players who all had QH ACL reconstruction. None were performed using BTB and thus selection bias has been limited. Although retrospective in nature, adequate records were obtained of all primary and secondary outcomes in all but one athlete over a 17 year period. Last, all surgeries were performed by the same individual thereby reducing the potential for small surgical differences impacting the reconstruction success rates.

This study demonstrated that 83% of players with eligibility remaining returned to play following ACL reconstruction and participated in 79.5% of the games remaining in their collegiate career. Kamath et al studied the incidence of re-injury and remaining eligibility used following ACL reconstruction in NCAA division I athletes across all sports. They examined players who underwent reconstruction during their college career (intercollegiate or IC) and compared them to those who had reconstruction prior to college (precollegiate or PC). They had 1 ACL re-injury of 54 in their IC group and 6 of 35 in their PC group. Six of these re-injuries were BTB autograft. The characteristics of the other failed graft were not noted. They also compared the percent eligibility used after ACL reconstruction, the methods of which were used as the model for this study. Players sustaining a PC injury used 78% of their total remaining eligibility, while players in their IC cohort used 77%. They defined RTP as return to a varsity roster.[18] Our study used a definition mirroring Shah et al in that successful return was defined as game play in an official NCAA or professional game after ACL reconstruction.[5]

The range of graft diameter in this study was 8.0mm to 9.5mm. A recent publication by Snaedbjornsson et al examined the incidence of re-injury within the first 24 months after ACL reconstruction. They demonstrated a decreased risk with QH vs BTB if the graft diameter was ≥ 9.0mm. They also demonstrated that likelihood of

re-injury was 0.84 times lower for each 1-mm increase in QH graft diameter. Re-injury risk was elevated with QH grafts <8mm in diameter. Their overall incidence of re-injury was 2.08% for QH and 2.63% for BTB.[19]

This cohort's average RTP was 318 days (~10.5 months). Schrock et al found in 2008 that 49% of NFL team physicians stated they would allow RTP after ACL reconstruction by 6 months, as compared with 14% of physicians in 2016. More commonly in 2016, RTP was delayed to 7 to 9 months.[4] Erickson et al, found that 55% of physicians allowed RTP at a minimum of 6 months.[20] Carver et al reported that in 2008, 70% of NCAA Division I football team physicians surveyed allowed RTP by 6 months. More recently in 2016, the majority of team physicians (51%) required 7 months or longer before allowing RTP which is a similar trend found in these results.[6]

Multiple studies have directly compared BTB and QH. Mohtadi et al performed a randomized clinical trial comparing BTB, QH and double bundle repair. They found a statistically significant higher rate of traumatic graft re-injury (partial or complete tears) in the QH and double bundle repair when compared to the BTB.[21] Sajovic et al randomized 56 patients to either BTB or QH and demonstrated no difference in Lysholm scores or KT-2000 anterior laxity at 5 and 11 year follow-up. However, at 17 years out in the same cohort, there was a statistically significant increase in anterior laxity measured by KT-2000 in the QH group versus the BTB group.[22] Nine players in the current study (26.5%) did require a secondary procedure on the ipsilateral knee, including revisions, chondroplasties and meniscal debridements, which could be related to the residual laxity of the graft.

Limitations

There are several weaknesses to this study, as it is retrospective and without a control arm or contrasting treatment arm, therefore there is no direct comparison of QH to BTB within this cohort. This sample is very heterogeneous with respect to position. Due to its small size, the study was underpowered to detect any statistical significance between variables. Additionally, the patient reported outcomes were recorded from a survey administered at different time points in patient recovery rather than a standardized period after surgery. These results are also limited to division I football players at a single institution and may not be generalizable to athletes of all levels.

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Conclusions:

QH demonstrated an ACL re-injury rate and RTP similar to previously published predominantly BTB ACL re-injury data in elite athletes. This study demonstrates that QH autograft may be a viable option in elite athletes.

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